

AIM:

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

CODE:

```
MINGW64:/d/OS_CM24112/OSLab_CM24112/Practical_10
GNU nano 8.5 sstf.c Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int visited[8] = {0};
    int head = 53, total = 0;

    for(int i = 0; i < 8; i++) {
        int min = 9999, index = -1;

        for(int j = 0; j < 8; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);
                if(dist < min) {
                    min = dist;
                    index = j;
                }
            }
        }

        total += min;
        head = req[index];
        visited[index] = 1;
    }

    printf("Total Head Movement = %d\n", total);
    return 0;
}
```

OUTPUT:

```
shars@shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practical_10 (main)
$ nano sstf.c

shars@shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practical_10 (main)
$ gcc sstf.c -o sstf

shars@shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practical_10 (main)
$ ./sstf
Sequence of Head Movement:
53 -> 41 (distance = 12)
41 -> 65 (distance = 24)
65 -> 67 (distance = 2)
67 -> 98 (distance = 31)
98 -> 122 (distance = 24)
122 -> 124 (distance = 2)
124 -> 183 (distance = 59)
183 -> 14 (distance = 169)

Total Head Movement = 323

shars@shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practical_10 (main)
$ |
```